Restriction enzymes

Mohamed Enan

The Nobel Prize in Physiology or Medicine 1978

for the discovery of restriction enzymes and their application to problems of molecular genetics"



Werner Arber
Switzerland



Hamilton Smith
USA



Daniel Nathans
USA

Nucleases

Nucleases are enzymes which hydrolyze nucleic acids.

Two types:

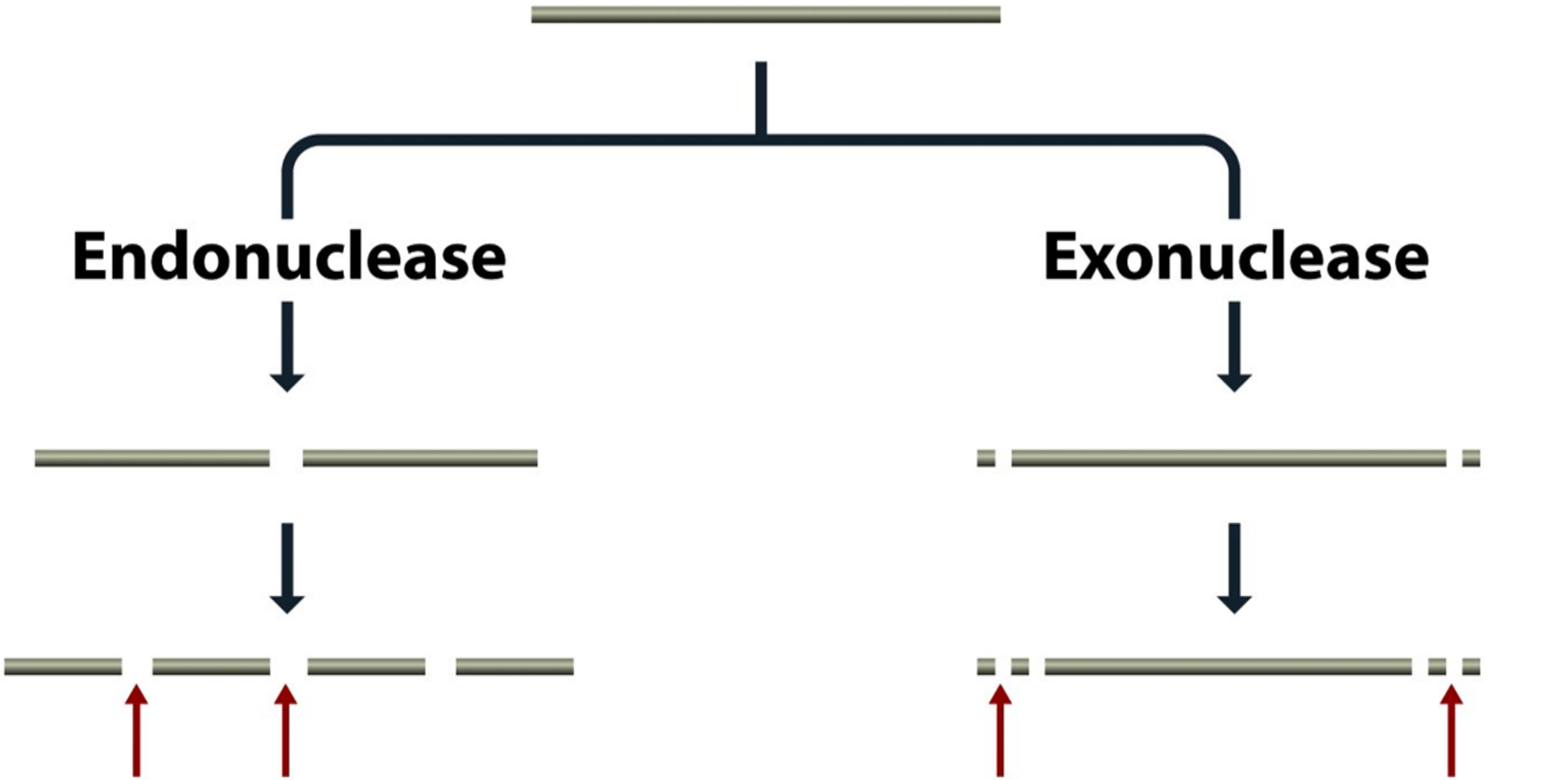
Endonuclease:

They are nuclease enzymes that cleave the phosphodiester bond in the polynucleotide from the interior.

Exonuclease:

They are nuclease enzymes that cleave the phosphodiester bond in the polynucleotide from the exterior.

Nucleases


Endonuclease
Exonuclease
Internal cuts
Nucleotides removed from the ends

What are restriction enzymes?

Restriction Endonucleases, these enzymes recognize and cut specific sequences in double-stranded DNA.

What are the Recognition site?

What are the Restriction Fragments?

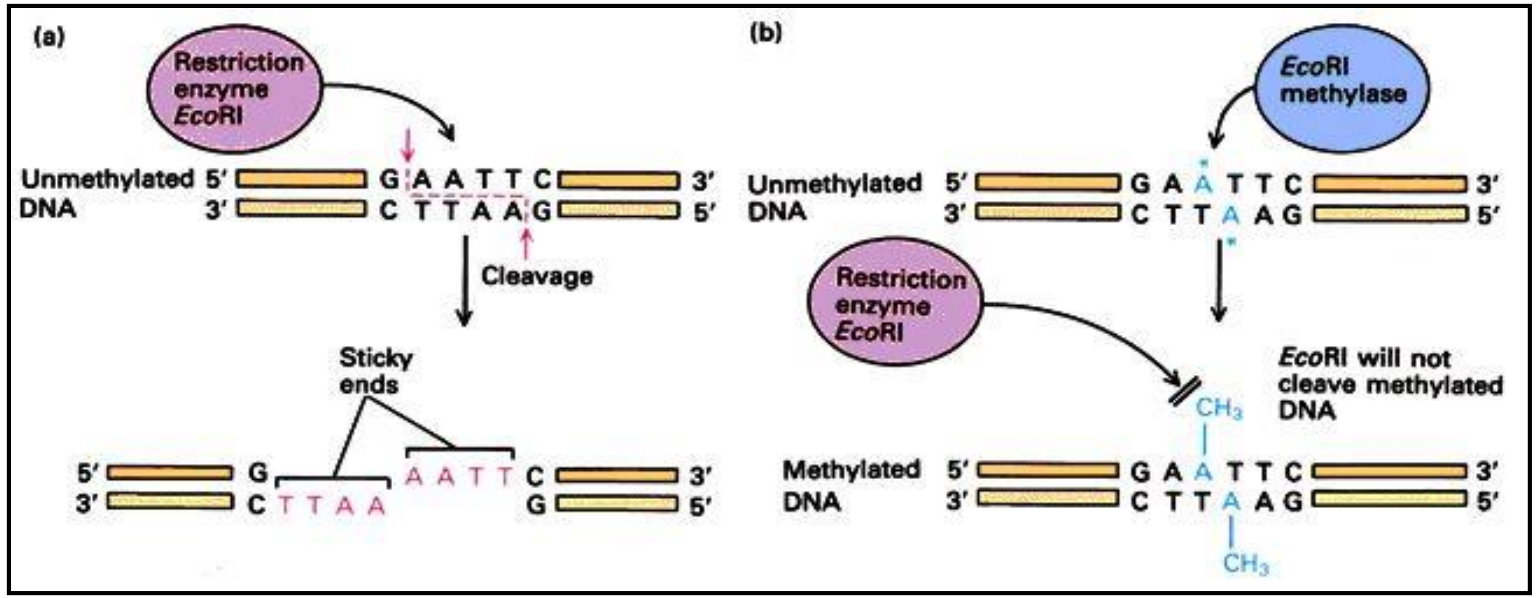
Function of REs

- **Restriction Enzymes are made to protect the bacteria from foreign DNA.**
- **Bacteria have a method of marking their own DNA as being "self" (called a Modification System).**
- **Any DNA not recognized as self is digested into smaller pieces by the Restriction Enzymes**

Restriction Endonucleases

- Called "restriction enzymes“ because restrict host range for certain bacteriophage
- Bacterial" immune system": destroy any "non-self" DNA

- Methylase recognizes same sequence in host DNA and protects it by methylating it; restriction enzyme destroys unprotected/non-self DNA. (restriction/modification systems)



Types

Naturally occurring restriction endonucleases are categorized into three groups (Types I, II III) based on their recognition sequence, subunit composition, cleavage position, and cofactor requirements.

Types of Restriction enzymes

There are three types of restriction enzyme that differ in the way recognize their restriction sites and where they cut the DNA:

Type I: restriction enzymes cut DNA about 1000 nucleotides after the recognition site and requires ATP and (S-Adenosyl methionine: AdoMet) as enzyme cofactors to function.

Type II: restriction enzymes cut DNA at the recognition site.

Type III: restriction enzymes cut DNA about 25-27 base pairs after the recognition site and requires ATP.

	Cleavage site	Location of methylase	Examples
Type I	Random Around 1000bp away from recognition site	Endonuclease and methylase located on a single protein molecule	EcoK I EcoA I CfrA I
Type II	Specific Within the recognition site	Endonuclease and methylase are separate entities	EcoR I BamH I Hind III
Type III	Random 24-26 bp away from recognition site	Endonuclease and methylase located on a single protein molecule	EcoP I Hinf III EcoP15 I

Naming of Restriction enzymes

Three-letter system is used.

The first three letters identify the organism from which the enzyme was isolated.

The first letter is from the Genus.

The next two are from the species name.

The names are pronounced in ways that follow no set pattern. e.g *EcoRI*, *HindIII*

Naming of REs

Restriction enzymes are named based on the bacteria in which they are isolated in the following manner:

E ***Escherichia*** ⟶ **Genus**

co ***coli*** ⟶ **species**

R **RY13** ⟶ **strain**

I **First identified** ⟶ **Order ID'd in bacterium**

REs	Source
HsaI	Homo sapiens
BamHI	Bacillus amylophilus
HindIII	Haemophilus influenzae
KpnI	Klebsilla pneumonia
TaqI	Thermus aquaticus
Sau 3A	Staphylococcus aureus.
Sma I	Serratia marcescens

What is a Palindrome?

A palindrome is anything that reads the same forwards and backwards:

English palindromes:

- **Mom**
- **Dad**

What are the palindrome sequence?

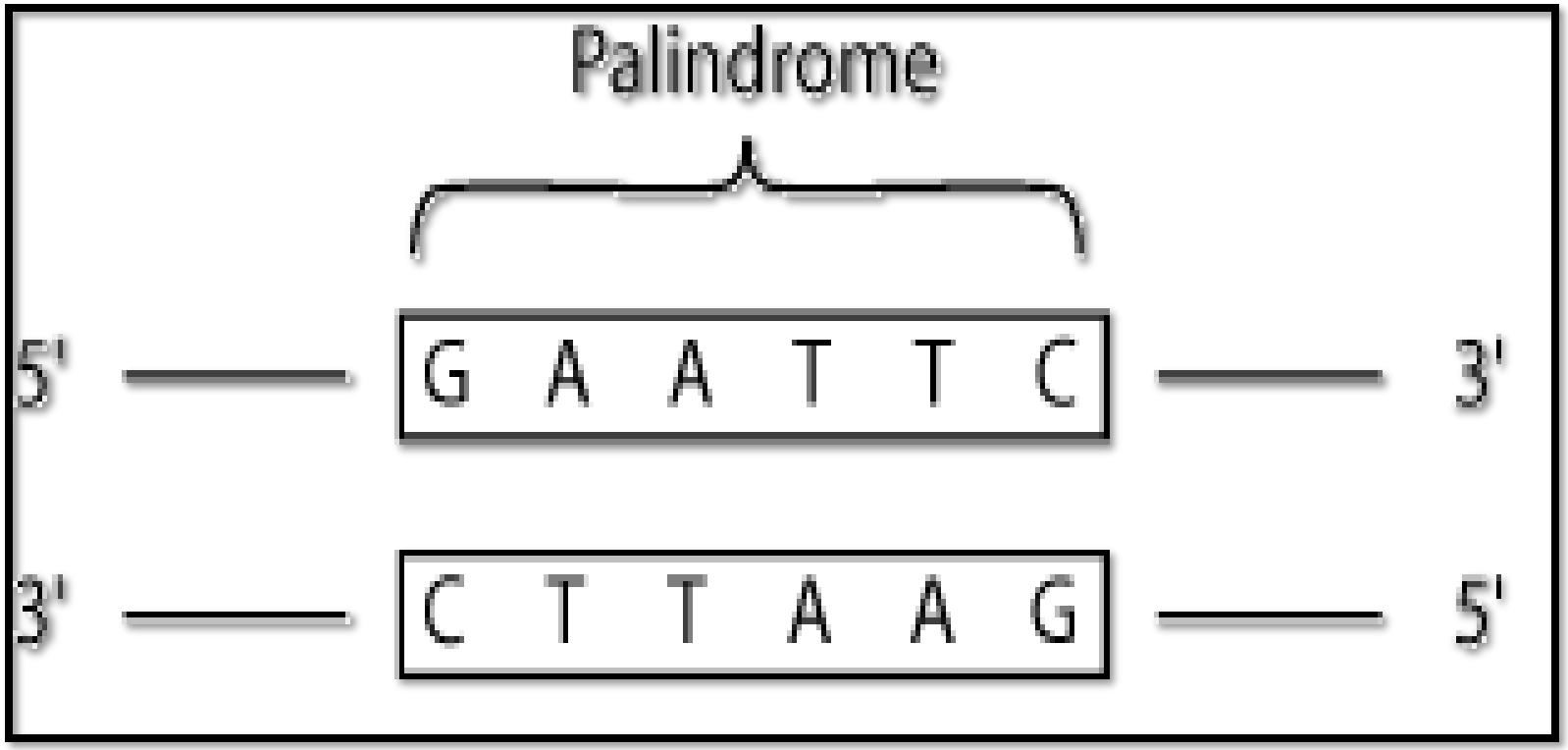
A palindrome is a sequence which is read the same on both strands in the 5' --> 3' direction.

5’-GAATTC-3’

3’-CTTAAG-5’

Palindromic sequence

A character sequence that reads the same both forward and backward is called a palindromic sequence.



Ways to cut DNA

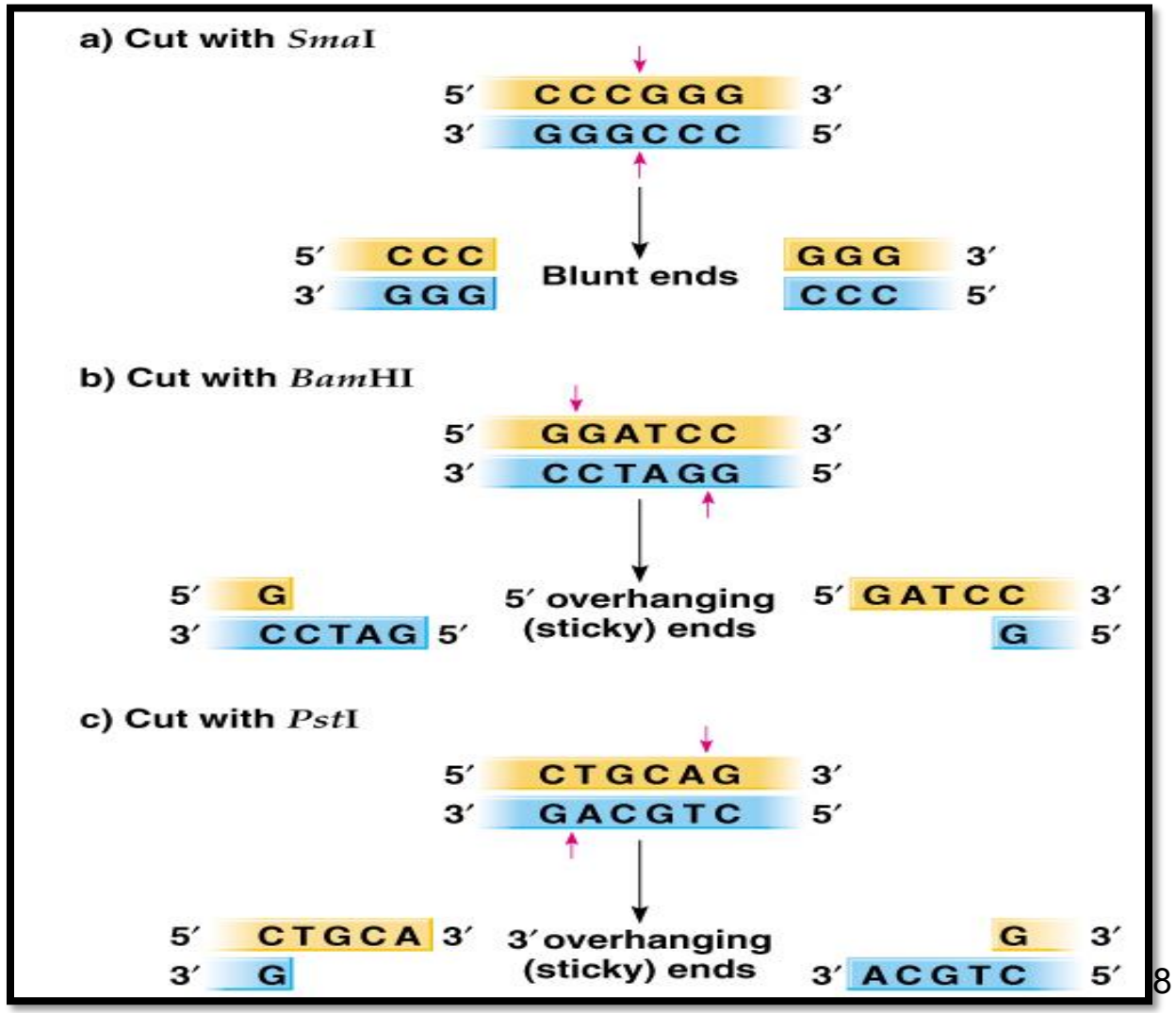
Sticky ends

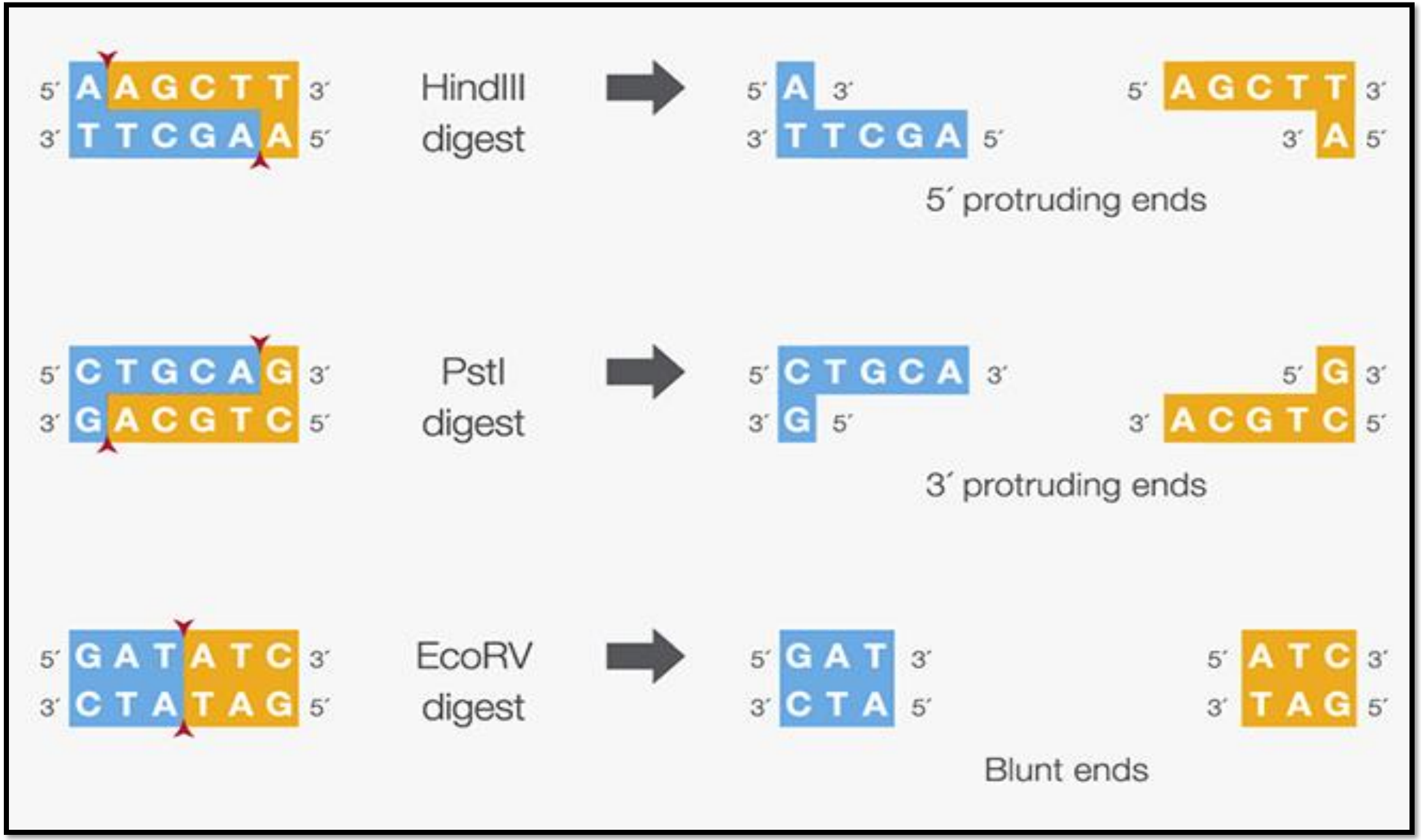
Restriction Enzymes can also produce fragments with single-stranded ends.

Blunt ends

Restriction Enzymes can also produce fragments without single-stranded ends.

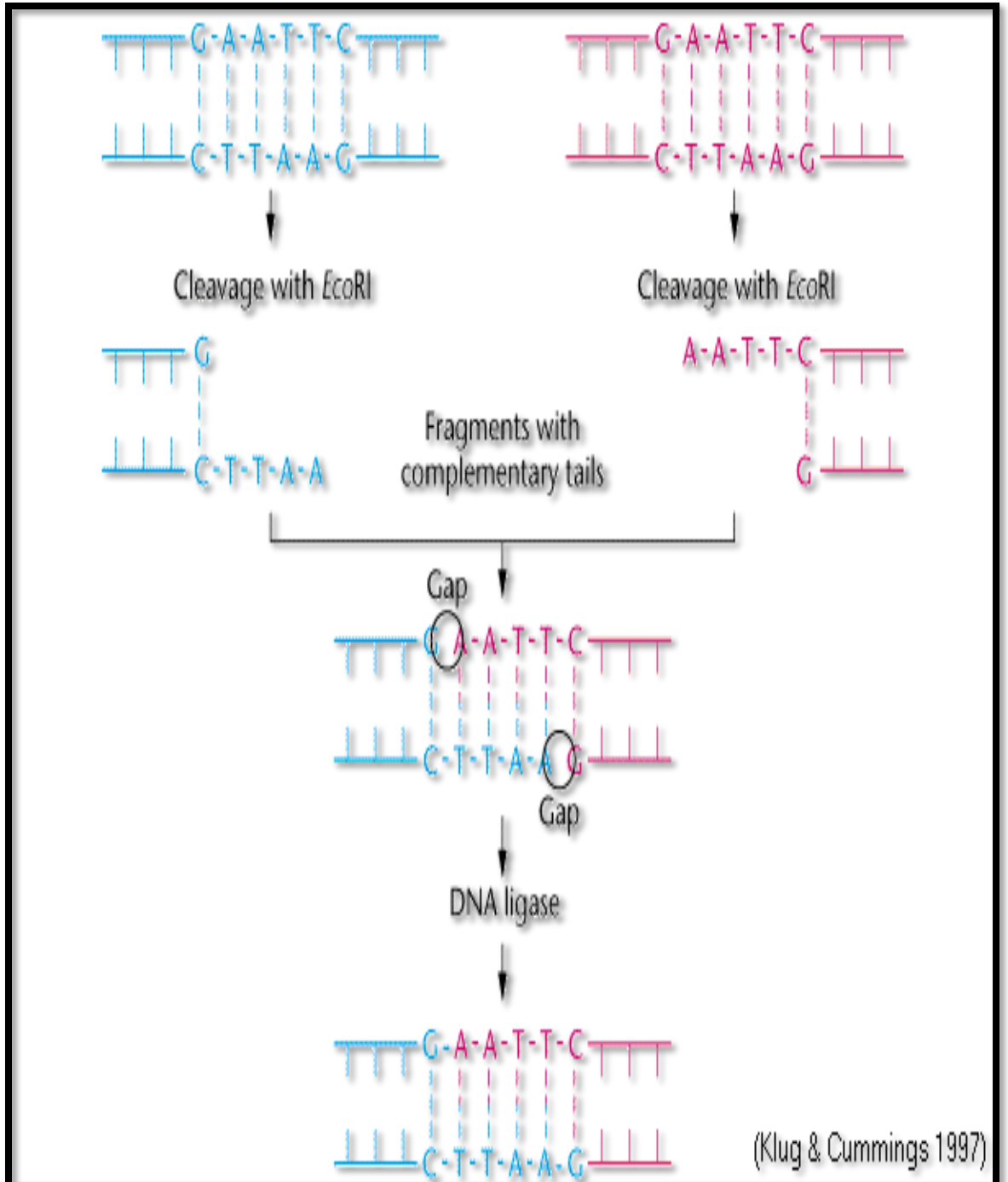
Sticky ends are useful for DNA cloning because complementary sequences anneal and can be joined directly by DNA ligase.




5′ A A G C T T 3′
3′ T T C G A A 5′
HindIII digest
5′ A 3′
3′ T T C G A 5′
5′ A G C T T 3′
3′ A 5′
5′ protruding ends
5′ C T G C A G 3′
3′ G A C G T C 5′
PstI digest
5′ C T G C A 3′
3′ G 5′
5′ G 3′
3′ A C G T C 5′
3′ protruding ends
5′ G A T A T C 3′
3′ C T A T A G 5′
EcoRV digest
5′ G A T 3′
3′ C T A 5′
5′ A T C 3′
3′ T A G 5′
Blunt ends

Ligation of Sticky ends

The single-stranded overhang from one fragment can be paired with any other fragment of DNA, including one from a different organism, that has a complementary sequence.



Isochozimers and non isoschizomers

Isoschizomer:

Enzymes that recognize and cut the same R. site e.g. DraI and Aha III/ Cfo I and HhaI.

Non-isoschizomer:

Enzyme that recognize the same DNA sequence but cut at different bases relatively to the R. site.

Isoschizomers









Neoschizomers









Isocaudomers

Enzymes that are able to recognize different recognition sites and generate the same ends.

EcoRI G|AATTC

MfeI C|AATTC

Produce 5' overhang AATTC

Sau3AI ...|GATC...

BamHI G|GATCC

Star Activity

5’-GAATTC-3’

3’-NTTAAG-5’

Conditions that contribute to star activity

1. High glycerol concentration > 5%
2. High unit to ug of DNA ratio
3. High pH
4. Presence of organic solvents.

Components of a restriction digest

1. DNA
2. Buffer
3. Enzyme

Restriction digestion reaction

Buffer volume

1/10 digest volume of 10X restriction buffer.

Enzyme volume:

The enzyme volume should not exceed 1/10 of the digest volume

One unit of activity

is typically defined as the amount of enzyme required to digest (or "restrict") one microgram of reference DNA in one hour at 37 °C.

Restriction digestion Reaction

1-Put the water and buffer into the tube first, then add the DNA.

2-Put the enzyme in last (avoid putting enzyme into water first, as it may start to break down). and mix by tapping the tube with your finger.

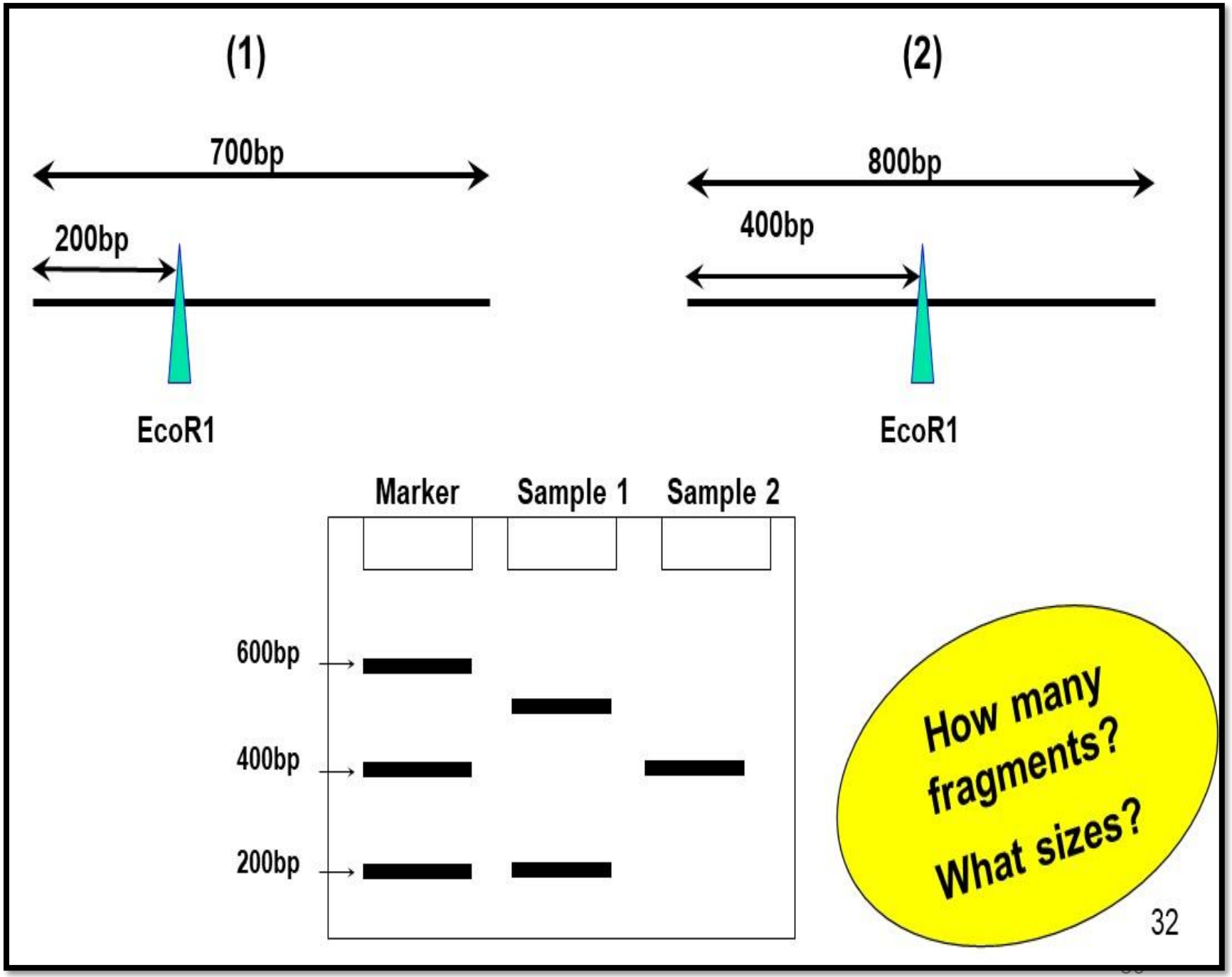
3.Quick spin to remove bubbles (DNA will adhere to bubble surface and becomes inaccessible to the enzyme).

4-Incubate the reaction in the 37ºC water-bath for a standard time of 90 minutes.

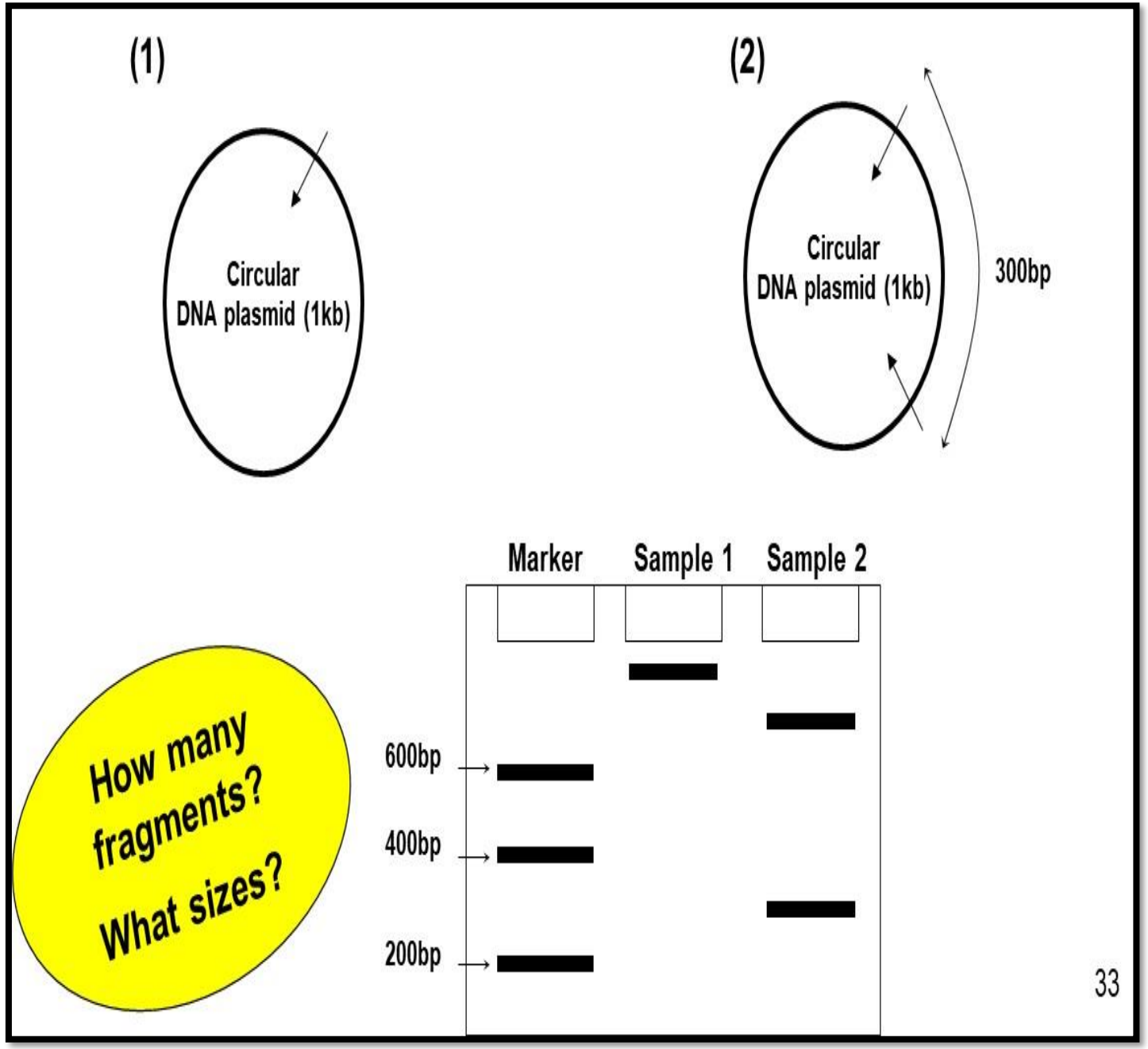
Minimum time for incubation= 5 minutes

Maximum time for incubation= over night

Examples

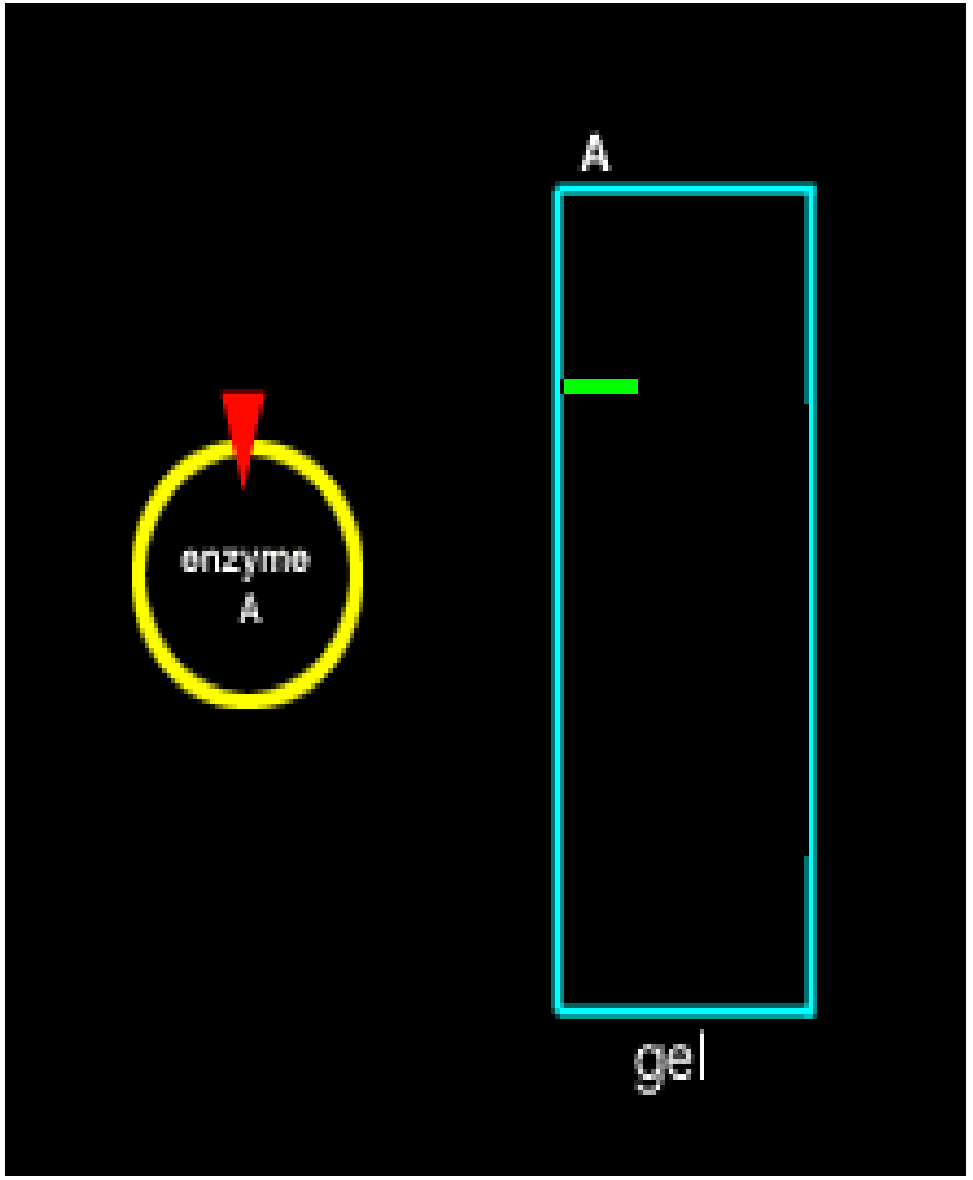

(1)
700bp
200bp
EcoR1
(2)
800bp
400bp
EcoR1
Marker
Sample 1
Sample 2
600bp
400bp
200bp
How many fragments?
What sizes?

Examples

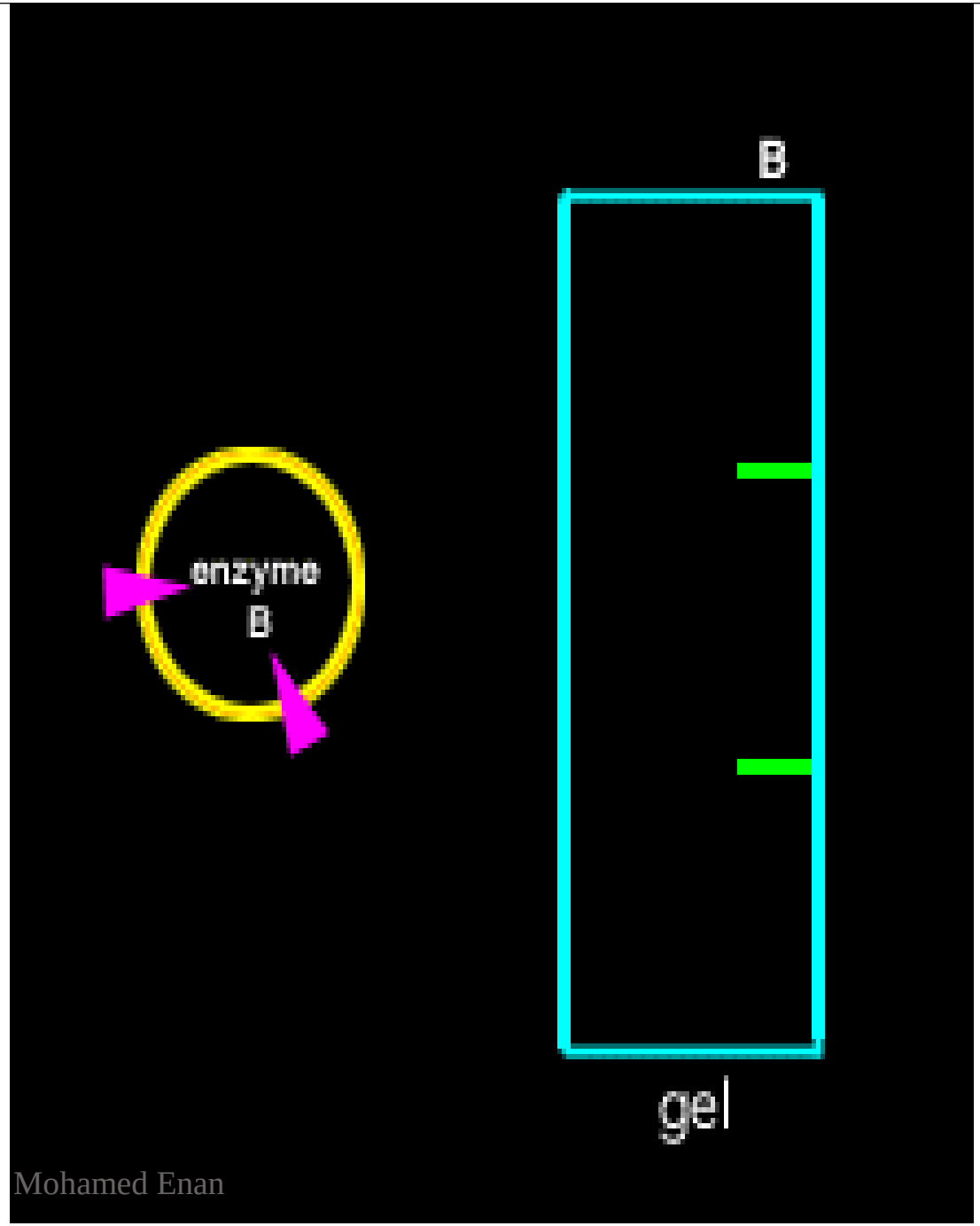

(1)
Circular
DNA plasmid (1kb)
(2)
Circular
DNA plasmid (1kb)
300bp
Marker
Sample 1
Sample 2
600bp
400bp
200bp
How many fragments?
What sizes?
33

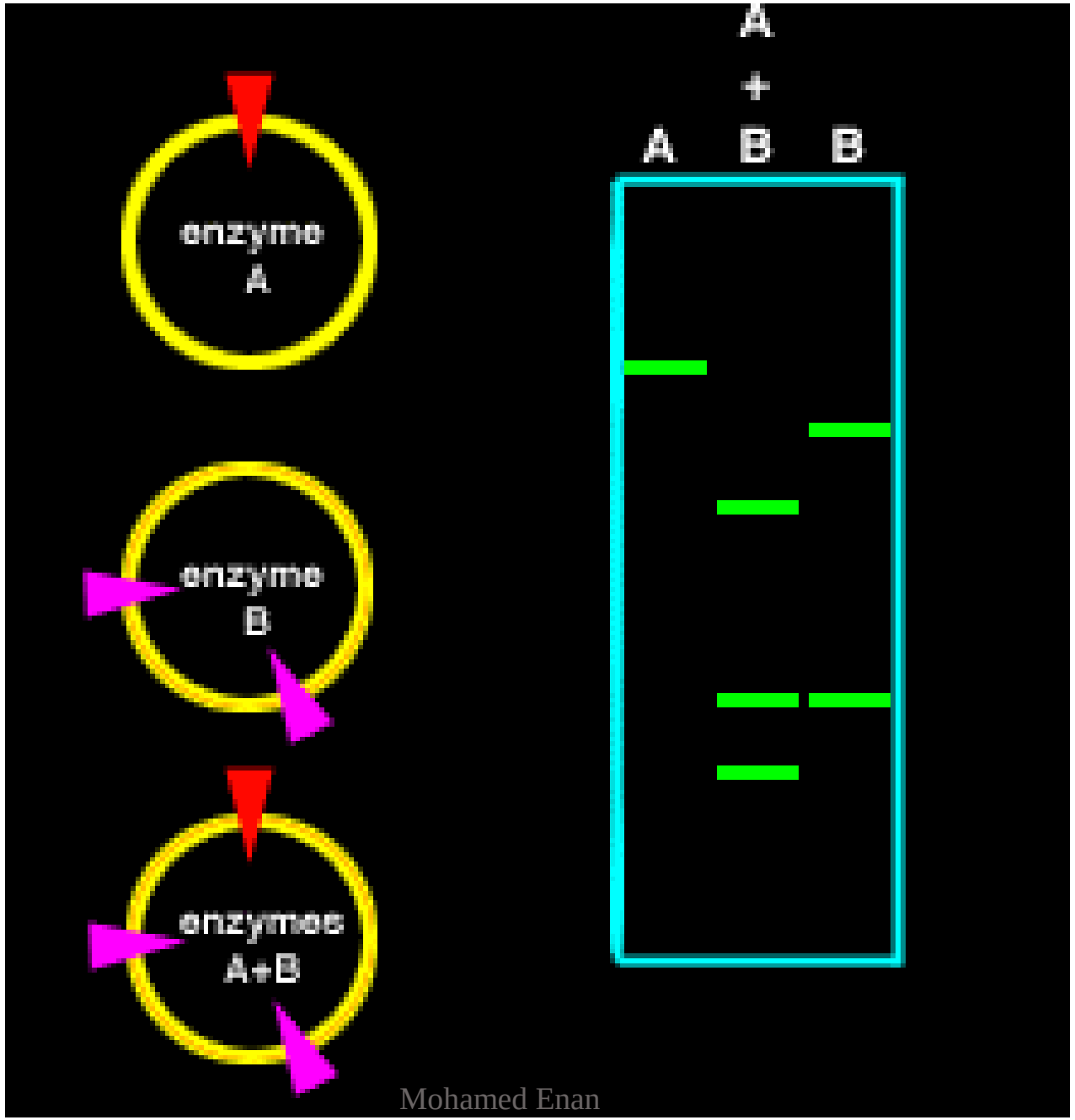
Restriction digestion of plasmid

Single digestion

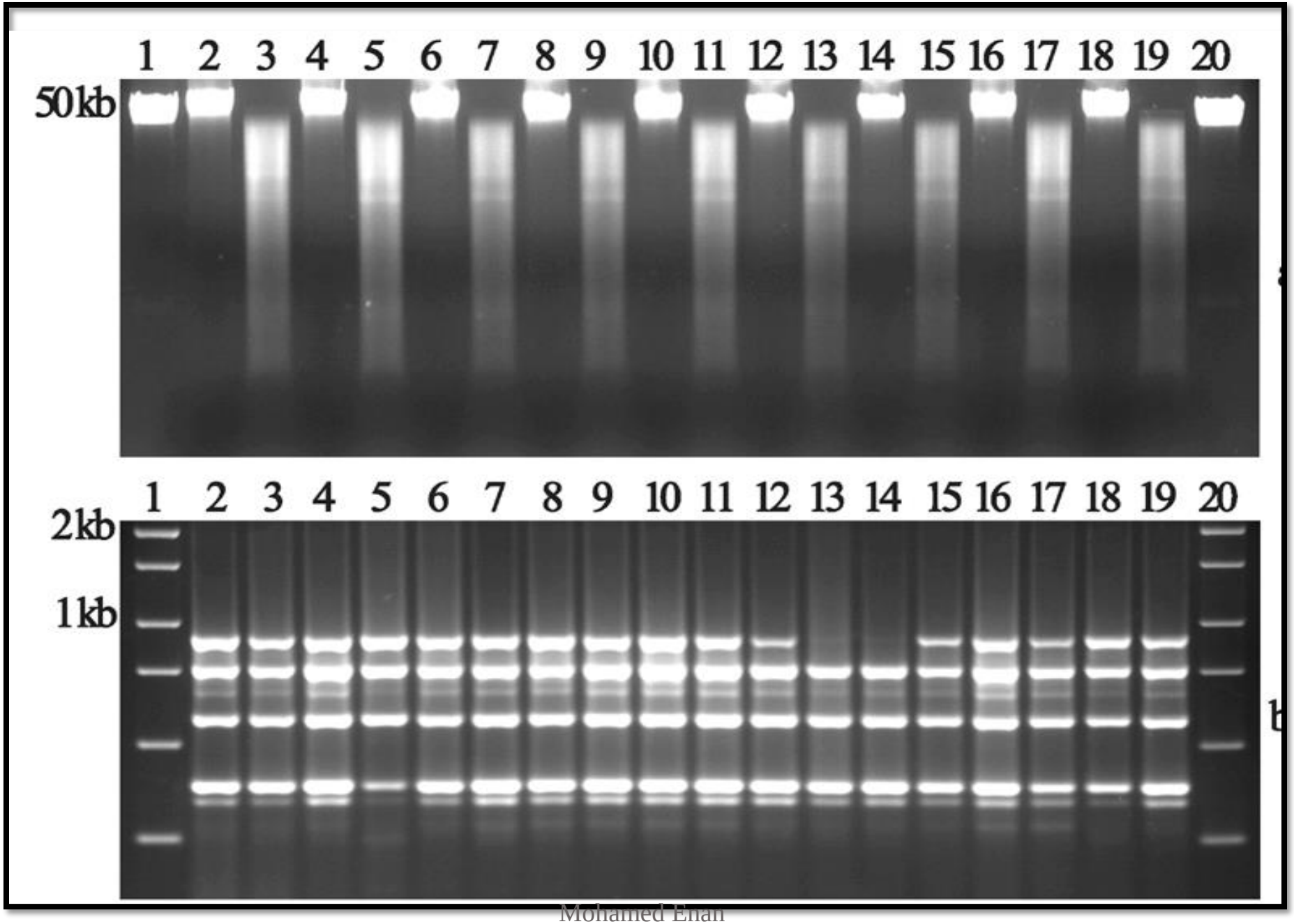


Double digestion





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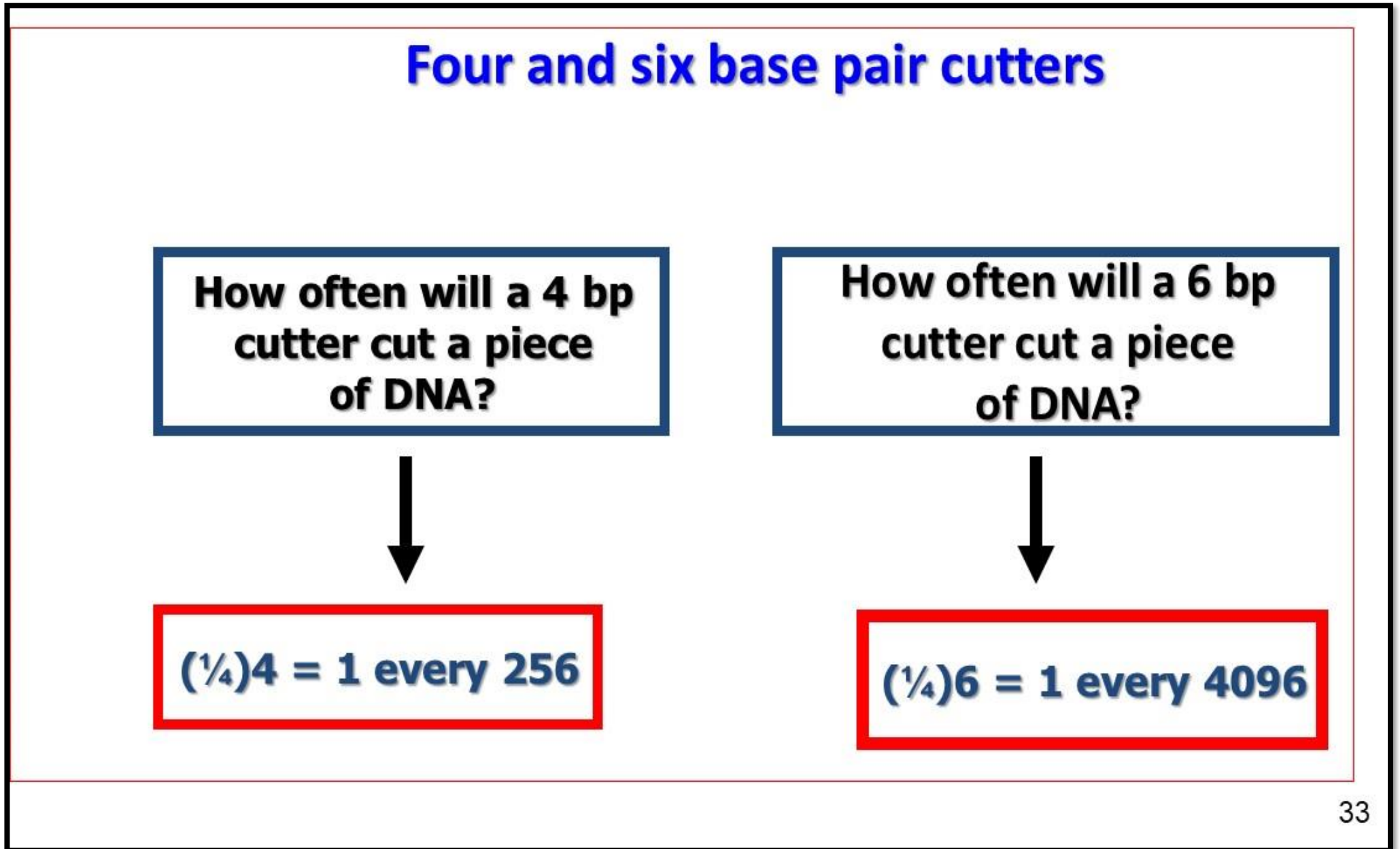
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How often does a R. Enzyme find the sequence it's looking for?

- It depends on how many bases it's looking for.
- There are 4 bases in any given strand of nucleic acid, right?
- That means that the probability of finding one of those, for example Adenine, at a particular position is 25%, or 1/4.
- If you are looking for an Adenine next to a Guanine, then the probability is 1/4 for each position, or

  1/4)x(1/4)=1/16.
- If you are looking for the string of bases AGC, the probability is (1/4)3=1/64.
- This means that every 64 bases, you would expect to find the string AGC.

How often does a R. enzyme find the sequence it's looking for?



How often does a R. Enzyme find the sequence it's looking for?

How often would you expect an enzyme like this to find what it's looking for?

(1/4)6 = 1/4096, or about once every 4,100 base pairs

But if you start with a linear gDNA that is 5,000 kb long (5 million base pairs) and digest it with a 6-cutter like *EcoRI*,

Statistically you would expect to end up with roughly 1221 individual fragments (5,000,000 divided by 4,096).

Applications of REs in Molecular Biology



With the aid of restriction enzymes, a particular gene can be removed from an organism's DNA. A genetically modified organism having a particular trait can then be produced by inserting the gene into an additional organism.

To examine how individual DNA varies, restriction fragment length polymorphism (RFLP) analysis can be utilized. Genetic variants can be found by cutting DNA with restriction enzymes and comparing the fragment sizes.

Restriction Mapping

A diagram that shows the location of restriction enzyme cut sites on a segment of DNA is known as a restriction map. Today, such maps would probably be made by sequencing the DNA and scanning the sequence for recognition sites. In the early days of molecular biology, restriction maps were derived experimentally. The DNA was digested with a series of restriction enzymes. The fragments of digested DNA were separated by agarose gel electrophoresis. This reveals how many recognition sites each enzyme has in the DNA and their distances apart

Restriction Mapping Example

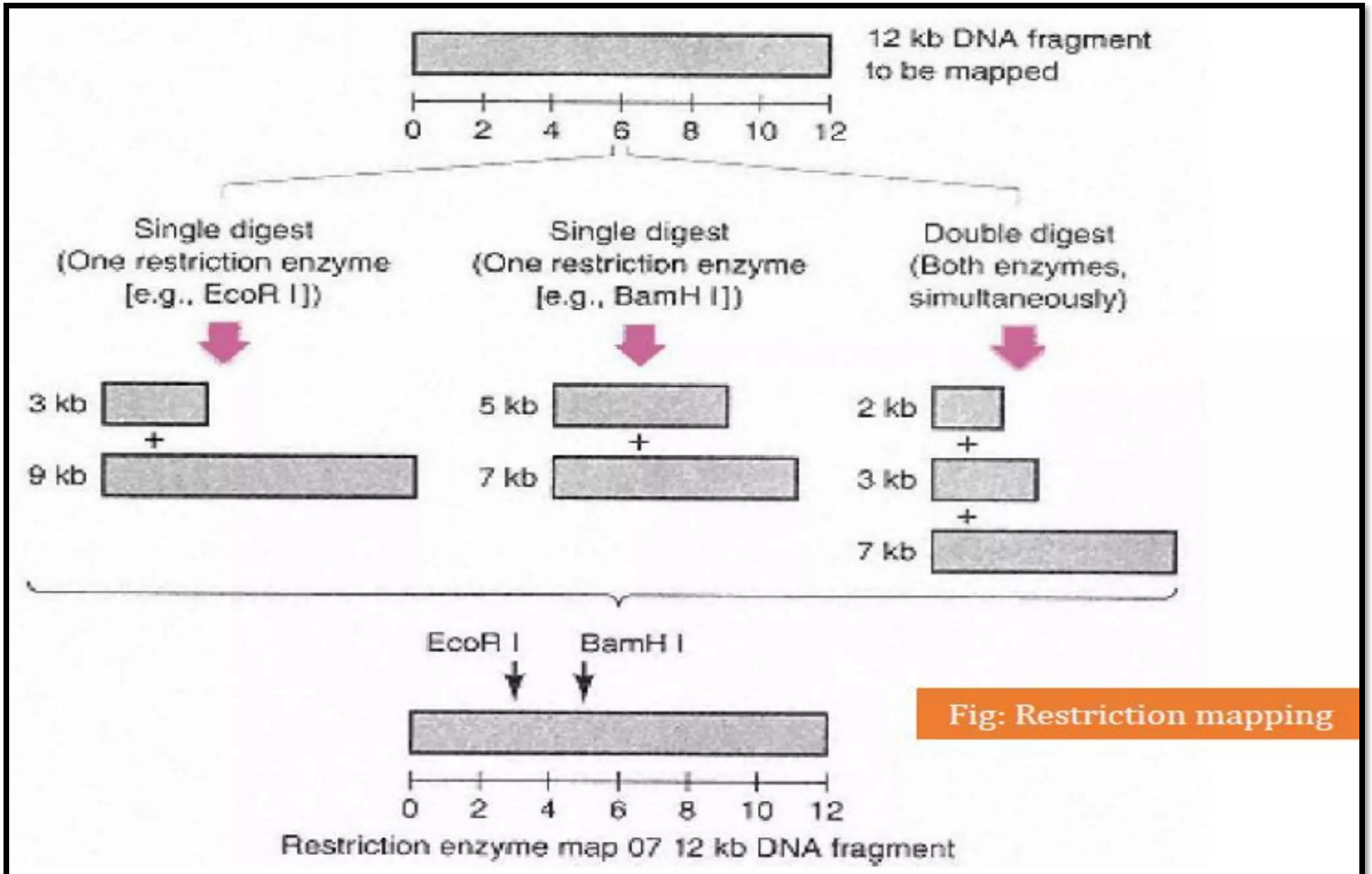
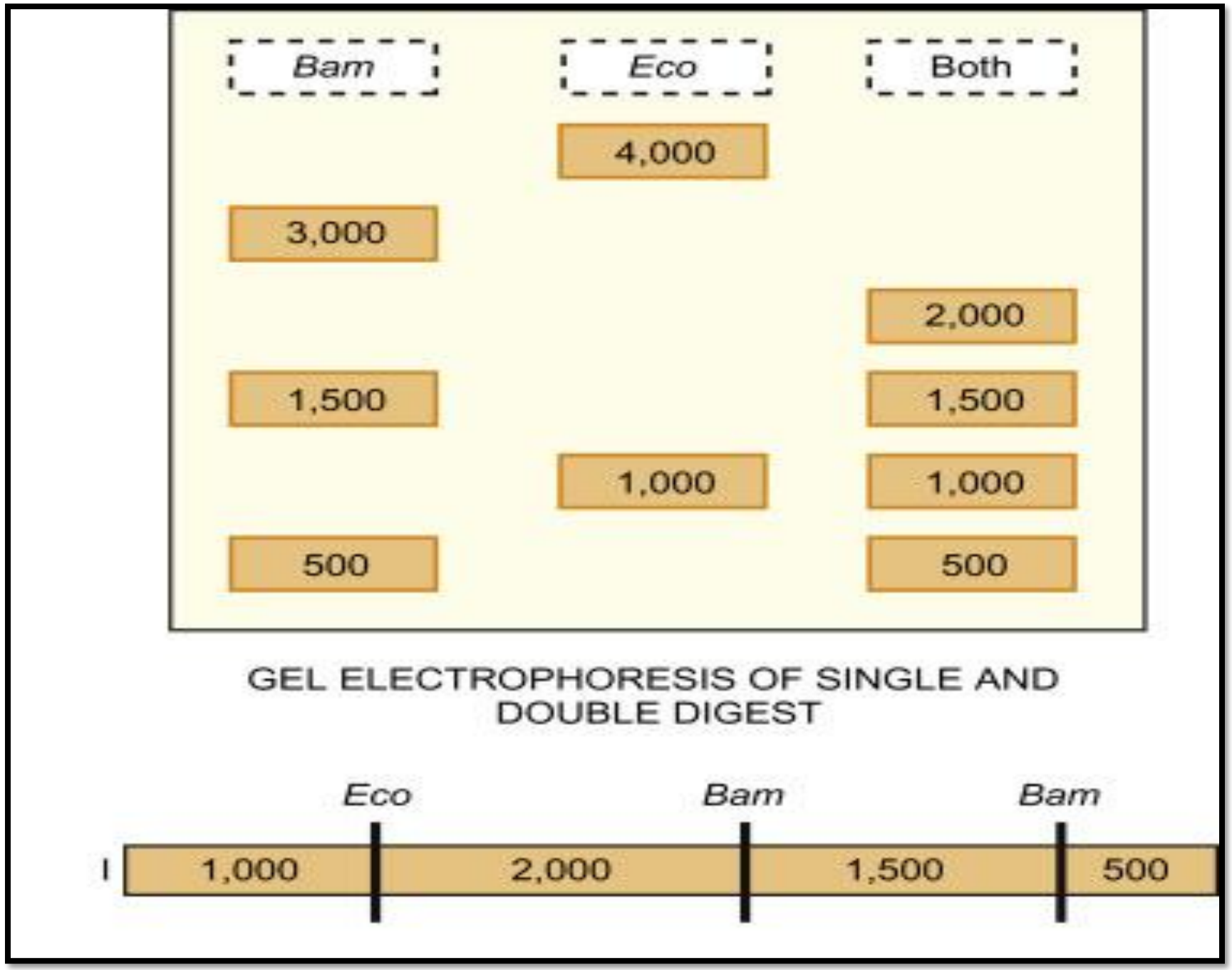


Fig: Restriction mapping



GEL ELECTROPHORESIS OF SINGLE AND DOUBLE DIGEST





The End